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Roll No. ....

**TMA-201**

**B. TECH. (SECOND SEMESTER)**

**MID SEMESTER**

**EXAMINATION, 2021-22**

**ENGINEERING MATHEMATICS—II**

**Time : 1½ Hours**

**Maximum Marks : 50**

**Note :** (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Solve : (CO1)

$$\frac{dy}{dx} = \sin(x + y) + \cos(x + y)$$

OR

(b) Solve : (CO1)

$$(1 + ae^{x/y})dx + ae^{x/y}\left(1 + \frac{x}{y}\right)dy = 0$$

2. (a) Solve : (CO1)

$$(D^4 - 1)y = e^x \cos x$$

**P. T. O.**

(2)

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OR

(b) Solve : (CO1)

$$x(x-1)\frac{dy}{dx} - (x-2)y = x^2(2x-1)$$

3. (a) Solve : (CO1)

$$\frac{d^3y}{dx^3} - 3\frac{d^2y}{dx^2} + 4\frac{dy}{dx} - 2y = e^x + \cos x$$

OR

(b) Solve  $\frac{d^2y}{dx^2} + y = \sec x$  by using variation of parameters method. (CO1)

4. (a) Find the Laplace transform of  $\frac{\cos at - \cos bt}{t}$ . (CO1)

OR

(b) Draw the graph of periodic function :

$$f(t) = \begin{cases} t & , \quad 0 < t < \pi \\ \pi - t & , \quad \pi < t < 2\pi \end{cases}$$

and find its Laplace transform. (CO2)

(3)

5. (a) Find inverse Laplace transform of

$$\frac{1}{s^2 - 5s + 6} \quad (\text{CO2})$$

OR

(b) Using Convolution Theorem, find : (CO2)

$$\mathcal{L}^{-1} \left\{ \frac{s^2}{(s^2 + a^2)(s^2 + b^2)} \right\}$$